

(a) Accuracy-driven forecasting

deep prediction of atmospheric fields

(b) Learned graph structure

prediction-optimized adjacency

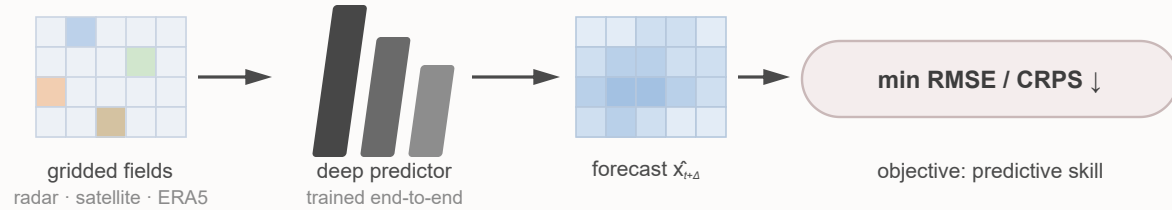
(c) Time-series causal discovery

lagged directed dependencies

(d-e) Regime-aware causal graphs

this work · Causal WeatherGraph

(a) Data-Driven Atmospheric Prediction



Nowcasting & video prediction

ConvLSTM ' 15 PredRNN ' 17 DGMR ' 21 Earthformer ' 22

Global & foundation-model forecasting

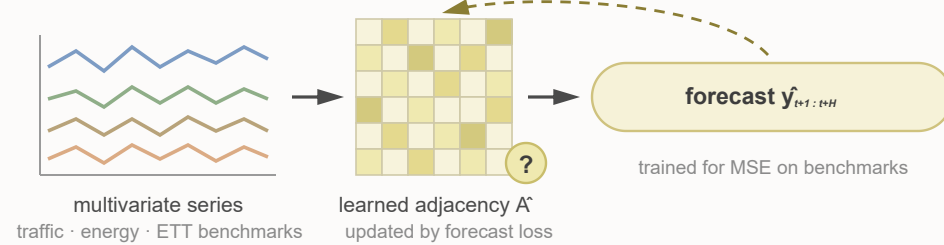
FourCastNet ' 22 Pangu-Weather ' 23 GraphCast ' 23 ClimaX ' 23 Stormer ' 24 NeuralGCM ' 24 GenCast ' 25 Aurora ' 25

domain journals: ALTF-Net ' 22 · DeepWind ' 24 (KBS)

✗ Not a causal readout

attention maps & internal graph weights are optimized for skill — they do not identify directed physical pathways

(b) Spatio-Temporal Graphs & Multivariate Time-Series Learning



Long-horizon multivariate forecasting

Informer ' 21 Autoformer ' 21 FEDformer ' 22 PatchTST ' 23 TimesNet ' 23 DLinear ' 23

Adaptive / evolving graph structure

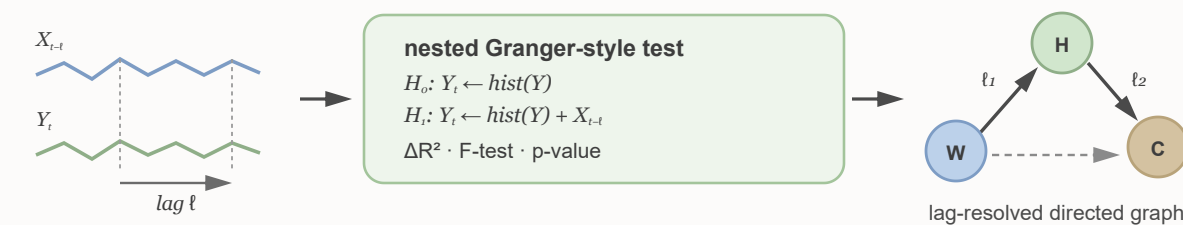
Adaptive STGNN ' 22 Dynamic ST graph ' 24 Adaptive graph gen. ' 25 Evolving GSL ' 25

representation & light backbones: TimeCLR ' 22 · lightweight MLP ' 24 (KBS)

✗ Prediction-optimized structure

learned adjacency is unconstrained & benchmark-driven — no physical admissibility, no significance / multiplicity control

(c) Causal Discovery for Earth-System Time Series



⚠ Hostile terrain: global reanalysis

strong autocorrelation · shared synoptic forcing
spherical spatial dependence · regime shifts
identifiability shown mainly on small systems

Foundations & Earth-system applications

Granger ' 69 Causal gateways ' 15 PCMCi ' 19 Causal networks ' 20 Causal review ' 23

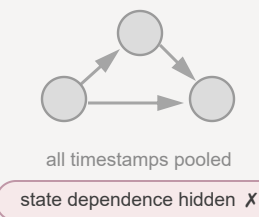
Neural & high-dimensional discovery

CUTS ' 23 Rhino ' 23 Non-stationary CSL ' 24 Partial-rank causal ' 24 Requested knowledge ' 25

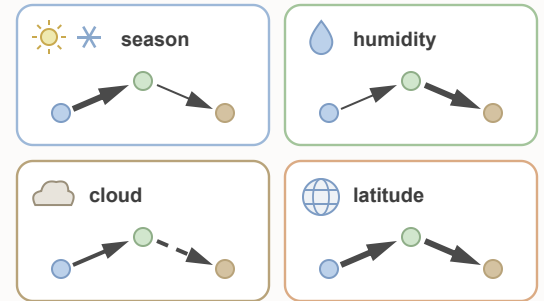
→ adopted as the screening backbone — constrained, corrected & falsified in this work

(d) Atmospheric Regimes & the Research Gap

single all-sample graph



VS



coupling strength, lag & spatial reach are state-dependent
(storm tracks · moisture transport · cloud–radiation feedback)

Research gap — no scalable framework combines:

- long-record global reanalysis
- physics-constrained regional graphs
- regime-specific estimation
- explicit layered validation chain

motivates

★ Causal WeatherGraph (Ours)

ERA5 global reanalysis · 1979–2025 · 6-hourly

$T = 68,668$ steps · $N = 2,048$ grid nodes · $V = 4$ variables

P, A

spatial aggregation → region–variable nodes

anomaly construction · scale normalization · regional series

C_{phys}

physics-guided candidate graph E^0

physically admissible variable & spatial relations only

$T_{lag} | s$

regime-wise lagged nested tests (7 regimes)

all-sample · DJF / JJA · high / normal humidity · high / normal cloud

V_{robust}

FDR control + layered validation chain

matched nulls · falsification · adjustment · PCMCi · bootstrap

stable regime-specific graphs $G_t^{(s)}$



lagged directed dependencies — not interventional causal claims